

1)

Eve can compute the first and third DHKA keys by exploiting the fact that the exponents used in each instance of the protocol are sequential. Let's call the random exponents used in the first DHKA instance as a_1 and b_1 , and the keys derived from the first, second and third DHKA instances as K_1 , K_2 , and K_3 respectively.

From the second DHKA, Eve has learned the key K_2 , which can be expressed as:

$$K_2 = g^{(a_1+1)(b_1+1)} \bmod p$$

We know that:

$$K_1 = g^{a_1 b_1} \bmod p$$

$$K_3 = g^{(a_1+2)(b_1+2)} \bmod p$$

$$K_1 = g^{a_1 b_1} \bmod p = g^{(a_1+1)b_1} / g \bmod p = (g^{(a_1+1)(b_1+1)} \bmod p) / g^{(b_1+1)} \bmod p = K_2 / g^{(b_1+1)} \bmod p$$

$$K_3 = g^{(a_1+2)(b_1+2)} \bmod p = g^{(a_1+1)(b_1+1)} g^2 / g^2 = K_2 g^2 / g^{(a_1+1)(b_1+1)} \bmod p$$

Therefore,

$$K_1 = K_2 / g^{(b_1+1)} \bmod p$$

$$K_3 = K_2 g^2 / g^{(a_1+1)(b_1+1)} \bmod p$$

2)

denote the discrete log of X with respect to base g as d , so we have:

$$X = g^d$$

We are given another value $X' \neq X$ and we can learn its discrete log, denoted as d' :

$$X' = g^{d'}$$

Our goal is to use this information to determine d .

We can start by taking the ratio of X and X' :

$$X / X' = (g^d) / (g^{d'}) = g^{(d - d')}$$

the ratio of X and X' is just a constant value that depends only on d and d' . we can take the discrete log of both sides with respect to base g to get:

$$\log_g(X / X') = \log_g(g^{(d - d')}) = d - d'$$

We know the values of X' and d' , so we can compute the left-hand side of the equation. Therefore, we can solve for d as follows:

$$d = d' + \log_g(X / X')$$

So we can use the given ability to learn the discrete log of X' to compute the discrete log of X .